

Roll No.

Total No. of Questions : 9]
(2042)

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**B.C.A. (CBCS) RUSA IInd Semester
Examination**

3745

DIGITAL ELECTRONICS

Paper : BCA-0203

Time : 3 Hours]

[Maximum Marks : 70

Note :- (i) Question No. 1 (Part-A) is compulsory. Attempt *four* questions choosing *one* question each from Part-B, C, D and E.

(ii) Figures at the right indicate marks.

Part-A

(Compulsory Question)

1. (A) Select the correct alternative for MCQs.

(i) With forward bias to a *pn* junction, the width of depletion layer :

(a) Increases

- (b) Decreases
 - (c) Remains the same
 - (d) None of these
- (ii) A digital circuit that can store only one bit is a :
- (a) Register
 - (b) NOR gate
 - (c) Flip-flop
 - (d) XOR gate
- (iii) The logical sum of two or more than two logical products is termed as :
- (a) OR operation
 - (b) POS
 - (c) SOP
 - (d) NAND operation

(iv) Which of the given logic family provide minimum power dissipation ?

- (a) JFET
- (b) CMOS
- (c) ECL
- (d) TTL

(v) The number of inputs in a half adder is :

- (a) 8
- (b) 2
- (c) 11
- (d) 32



(vi) What is the value to be considered for a 'don't care condition' ?

- (a) 0
- (b) 1
- (c) Either 0 or 1
- (d) Any number except 0 and 1

(vii) Which device has one input and many outputs ?

- (a) Multiplexer
- (b) Demultiplexer
- (c) Counter
- (d) Flip-flop

(viii) A K-map of n -variables contains cells.

(ix) The condition $S = R = 1$ is called as condition.

(x) CMOS stands for $1 \times 10 = 10$

(B) Answer the following in **25** to **50** words :

(i) Describe combinational circuit.

(ii) Distinguish between TTL and CMOS families.

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(iii) Draw symbol and truth table of OR, NOT and NAND gate.

(iv) Explain the function of J-K Flip-flop.

(v) Draw the circuit diagram of an 8-input multiplexer.

$4 \times 5 = 20$

Part-B

(Unit-I)

10 each

2. Discuss in detail Bipolar Junction Transistor and draw its circuit symbol.

3. Explain the non-saturated bipolar logic family, ECL in detail.

(Unit-II)

4. Explain the various laws of Boolean algebra. Also state De-Morgan's theorem with example. 10
5. (a) Discuss 'NAND gates are Universal Gate'.
 (b) Draw the circuit diagram of NOR gate and also give its truth table. 6,4

Part-D**(Unit-III)**

10 each

6. Simplify the following Boolean function using K-map and draw the circuit for simplified expression :

$$F(W, X, Y, Z) = \Sigma(0, 2, 4, 5, 9,$$

11, 14, 15)

7. Explain the SOP form and POS form of simplifying Boolean expression using K-maps.

Part-E**(Unit-IV)**

10 each

8. What is a Flip-flop ? Compare the operations of D and T Flip-flops with the help of their truth-table.
9. Draw and explain the working of Full Adder Circuit with Truth-table